Kerala Polytechnic Syllabus Revision 2026

Materials and Measurements Laboratory

Course Code: 2028 **Semester: II** **Credits: 1.5**

Dept: Mechanical Engineering

A hands-on practical course focusing on material identification, microstructure analysis, heat treatment, non-destructive testing, and precision metrology.

Download Study Notes

Official Source Declaration

This lesson is architected based on the official **SITTTR Kerala Revision 2026** syllabus for Course Code **2028 (Materials and Measurements Laboratory)**.

- **Authoritative Source:** SITTTR Syllabus PDF (5 Pages).
- **Curriculum:** Revision 2026.
- **Accuracy:** All module hours, course outcomes, and assessment schemes are extracted directly from the official document.

Course Dashboard

Teaching Scheme

Component	Value
Practical (P) Hours/Week	3 Hours
Self-Learning (SS) Hours/Week	1.5 Hours

Total Contact Hours/Semester	45 Hours
Total Credits	1.5

Examination Scheme

Assessment	Marks
Continuous Internal Evaluation (CIE)	60
End Semester Examination (ESE)	40
Total Marks	100

How to Study this Lab Course

This is a practical-heavy course. To excel:

1. **Pre-Lab:** Read the theory behind each instrument (e.g., Vernier principle) before entering the lab.
2. **Precision:** In metrology (Module 3 & 4), accuracy is everything. Always check for zero errors.
3. **Observation:** For NDT and Microstructure, develop a keen eye for patterns and defects.
4. **Documentation:** Maintain your record book neatly; it carries 4 marks in ESE and significant weight in CIE.

Course Objectives & Outcomes

Course Objectives

- Familiarize properties, microstructure, and classification of engineering materials.
- Understand heat treatment effects and NDT methods.
- Perform accurate linear and angular measurements.
- Apply metrology principles for threads and precision components.

Course Outcomes (CO)

CO	Description	Bloom's Level
CO1	Apply material identification techniques to distinguish materials and study microstructure.	Apply
CO2	Apply non-destructive testing techniques (Visual & Dye Penetrant).	Apply
CO3	Apply linear measurement techniques (Vernier, Micrometer, Height Gauge, Dial Indicator).	Apply
CO4	Apply angular and thread measurement techniques for precision applications.	Apply

Syllabus Coverage Map

Module	Focus Area	Experiments	Hours
Module I	Material Science Basics	Expt 1, 2	3
Module II	Heat Treatment & NDT	Expt 3, 4, 5, 6	15 (incl. Series I)
Module III	Linear Metrology	Expt 7, 8, 9, 10	12
Module IV	Angular & Thread Metrology	Expt 11, 12, 13, 14	15 (incl. Series II)

Learning Roadmap

Step 1: Material Identity

Learn to distinguish metals by looking at them, filing them, or checking magnetism.

Step 2: Internal Structure & Integrity

See how heat changes metals and how to find invisible cracks using NDT.

Step 3: Precision Linear Measurement

Master the tools that measure to the nearest 0.02mm or 0.01mm.

Step 4: Complex Geometry

Measure angles, tapers, and screw threads for industrial assembly.

Module I

Material Identification & Microstructure

Instructional Hours: 3   Mapped to: CO1

Expt 1: Study and Identification of Engineering Materials

Engineering materials are broadly classified into Ferrous (containing Iron) and Non-Ferrous. Identification is the first step in any workshop.

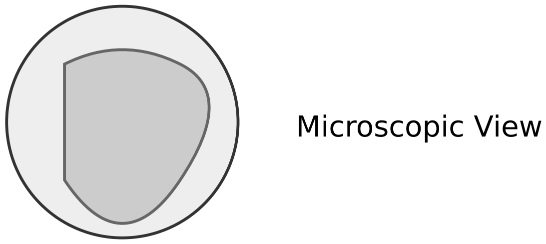
- **Appearance:** Color and texture (e.g., Copper is reddish, Cast Iron is dull grey).
- **Magnetism:** Ferrous materials (except some stainless steels) are magnetic.
- **Filing Test:** Resistance to a file indicates hardness.
- **Spark Test:** Touching the metal to a grinding wheel. The color, shape, and length of sparks identify the carbon content.

Malayalam support:

[illegible] [illegible] [illegible], [illegible] (Magnetism), [illegible] [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] (Spark) [illegible] [illegible] [illegible] [illegible].

Expt 2: Study of Microstructure

Using a metallurgical microscope to observe the grain structure of metals. Students use standard charts to identify phases like Ferrite, Pearlite, or Cementite.



Schematic of grain boundaries in a metal specimen.

Module II

Heat Treatment & Non-Destructive Testing

Instructional Hours: 15   Mapped to: CO2

Expt 3: Heat-Treated Specimens

Comparing specimens that have undergone:

- **Annealing:** Slow cooling for softness.
- **Normalizing:** Air cooling for grain refinement.
- **Hardening (Quenching):** Rapid cooling for maximum hardness.
- **Tempering:** Reheating to reduce brittleness.

Expt 4-6: Non-Destructive Testing (NDT)

NDT allows inspecting parts without damaging them.

- **Visual Inspection (VI):** Using magnifying glasses and mirrors to find surface flaws.
- **Dye Penetrant Test (DPT):**
  1. Clean surface.
  2. Apply red penetrant (wait for dwell time).
  3. Remove excess penetrant.
  4. Apply white developer. Cracks appear as bright red lines.
- **Magnetic Particle Test (MPT):** Using magnetic fields to find surface and near-surface cracks in ferrous metals.

Malayalam support:

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Module III

Linear Measurement

Instructional Hours: 12   Mapped to: CO3

Precision Instruments (Vernier & Micrometer)

Vernier Caliper

Least Count (LC) = 1 MSD - 1 VSD. Typically 0.02 mm.

Vernier Reading Calculator

Main Scale Reading (mm): 10

Vernier Scale Division (VSD): 5

Total Reading: 10.10 mm

Micrometer (Screw Gauge)

Works on the principle of screw and nut. Pitch / No. of divisions on circular scale. Typically 0.01 mm.

Malayalam support:

Vernier Caliper □□□□□□□□□□ □□□□ □□□□ (External), □□ □□□□ (Internal), □□□□□ (Depth) □□□□□□ □□□□□□□□□□ □□□□□□□. Micrometer □□□□□□ □□□□□□□□□□□ (0.01mm) □□□ □□□□□□□ □□□□□□□□□□□□□□□.

Module IV

Angular & Thread Measurement

Instructional Hours: 15 Mapped to: CO4

Expt 11-12: Angular & Taper Measurement

- **Bevel Protractor:** Measures angles with a vernier scale (Least count 1 minute).
- **Sine Bar:** Used with slip gauges for high-precision angle measurement based on trigonometry ($\sin \theta = h/L$).

Example: Sine Bar Calculation
Given: Length of Sine Bar (L) = 100 mm, Height of Slip Gauges (h) = 25.88 mm.
Required: Taper Angle (θ).
Calculation: $\sin \theta = h / L = 25.88 / 100 = 0.2588$.
$\theta = \sin^{-1}(0.2588) \approx 15°$.
Answer: 15 degrees.

Expt 13-14: Threads & Feeler Gauges

Thread Parameters: Measuring Major diameter, Minor diameter, and Pitch using a screw thread micrometer or wire method.

Feeler Gauge: A set of thin steel blades used to measure clearance or gap between mating parts (e.g., spark plug gap).

Student Activities for Self-Learning

Week	Activity Description	Hours
1-3	Watch videos on material classification; Prepare comparison tables.	4.5
4-8	Review heat treatment basics; Study NDT safety precautions.	7.5
9-12	Practice reading Vernier/Micrometer from virtual simulators.	6.0
13-15	Review thread parameters and angular measurement concepts.	4.5

Formula Bank

Vernier Caliper

LC = 1 MSD - 1 VSD

Total Reading = MSR + (VSD × LC)

Micrometer

LC = Pitch / No. of Circular Scale Divisions

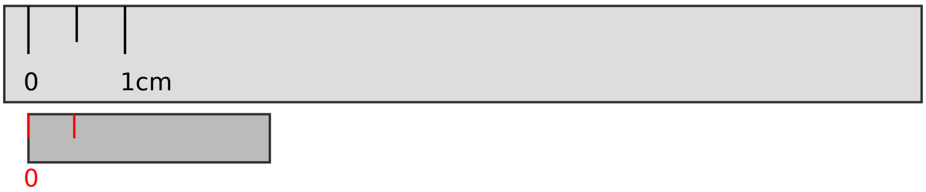
Total Reading = PSR + (HSR × LC)

Sine Bar

$\sin\theta = h / L$

(h = Height of slip gauges, L = Length of sine bar)

Visual Library



Simplified Vernier Caliper Scale Principle.

Assessment Methodology

Mode	Details	Marks
CIE - Attendance	Regularity in lab	5
CIE - Continuous Evaluation	Record, Procedure, Setup, Viva	30
CIE - Practical Tests	CA1 (7th week) & CA2 (14th week)	15
CIE - Open Ended Expt	Originality & Execution (15th week)	10
Total CIE		**60**
ESE (Practical)	3-hour external exam	**40**

Module-wise Questions & Answers

Module I & II

What is the difference between Ferrous and Non-Ferrous materials?

Ferrous materials contain iron as the main constituent (e.g., Mild Steel, Cast Iron) and are usually magnetic. Non-ferrous materials do not contain iron (e.g., Aluminum, Copper, Brass) and are non-magnetic.

Explain the purpose of the 'Developer' in Dye Penetrant Testing.

The developer is a white powder that acts like a blotter. It pulls the trapped penetrant out of the cracks to the surface, making the defect visible against the white background.

Module III & IV

Define Least Count of a measuring instrument.

The Least Count is the smallest value that can be measured accurately by a measuring instrument. For a standard Vernier Caliper, it is 0.02 mm.

Why is a Surface Plate made of Granite?

Granite is used because it is hard, wear-resistant, non-magnetic, and does not rust. It also has high thermal stability, meaning it doesn't expand or contract easily with temperature changes.

Practice Model Paper

(Not an Official Examination Paper)

Part A: Identification (10 Marks)
Identify the given 5 metal specimens using visual and physical tests.

Part B: Measurement (20 Marks)
1. Measure the internal diameter and depth of the given hollow cylinder using a Vernier Caliper.
2. Determine the taper angle of the given wedge using a Sine Bar and Slip Gauges.

Part C: Viva Voce (10 Marks)

Questions on NDT procedures and least count calculations.

Quick Revision

- **Spark Test:** Long yellow sparks = Low Carbon Steel; Short bushy sparks = High Carbon Steel.
- **DPT Steps:** Clean → Penetrant → Clean → Developer → Inspect.
- **Vernier LC:** 0.02 mm.
- **Micrometer LC:** 0.01 mm.
- **Sine Bar:** Always used on a Surface Plate.
- **Feeler Gauge:** Used for clearances, not for measuring diameters.

Glossary

- **Pitch**

Metrology	The science of measurement.
Parallax Error	Error caused by viewing the scale from an angle.
Quenching	Rapid cooling of a metal in water or oil to harden it.
The distance between two consecutive thread crests.	

References

- *Material Science and Engineering* - V. Raghavan (Prentice Hall).
- *Engineering Metrology* - Er. R.K. Jain (Khanna Publishers).
- *Engineering Metrology and Measurements* - Raghavendra and Krishnamurthy (Oxford).
- NPTEL Course on Metrology: Link